Dept. of Computer Engineering, NIT Kurukshetra

ITPC-205 (OOP using JAVA), Mid sem 1

Full Marks: 5*3=15

Total time: 50 mins

1. (Game: scissor, rock, paper) Write a program that plays the popular scissor-rock paper game. (A scissor can cut a paper, a rock can knock a scissor, and a paper can wrap a rock.) The program randomly generates a number 0, 1, or 2 representing scissors, rock, and paper. The program prompts the user to enter a number 0, 1, or 2 and displays a message indicating whether the user or the computer wins, loses, or draws.

```
scissor (0), rock (1), paper (2): 1 ↵Enter
The computer is scissor. You are rock. You won
```

2. Take a string from the user and check if the string is a valid string in **Binary** or not. Define a custom exception called *BinaryFormatException*. Implement a *bin2Dec* method to throw a *BinaryFormatException* if the string is not a binary string.

3. (Markov matrix) An n * n matrix is called a positive Markov matrix if each element is positive and the sum of the elements in each column is 1. Write the following method to check whether a matrix is a Markov matrix.

public static boolean isMarkovMatrix(double[][] m).

Write a test program that prompts the user to enter a 3 * 3 matrix of double values and tests whether it is a Markov matrix. Here is a sample run:

```
Enter a 3-by-3 matrix row by row:
0.15 0.875 0.375 ↵Enter
0.55 0.005 0.225 ↵Enter
0.30 0.12 0.4 ↵Enter
It is a Markov matrix
```